9

Open systems interconnection model

The purpose of the open systems interconnection reference model is to provide a common basis for the development of systems interconnection standards. An open system is a system that conforms to specifications and guidelines, which are 'open' to all.

Objectives

When you have completed studying this chapter you will be able to:

- Describe data communications for instrumentation and control
- List and describe the seven layers of the OSI reference model
- Give an example of an OSI model application

9.1 Data communications for instrumentation and control

In digital data communications wiring together two or more devices is one of the first steps in establishing a network. As well as this hardware requirement, software must also be addressed. Where all the devices on a network are from the same manufacturer, any hardware and software problems are usually overcome easily because the system is generally designed within the same guidelines and specifications. However, it is where devices from different manufacturers are used on the same network that the problems seem to begin.

Networks which use devices from one manufacturer and that work with specific hardware connections and protocols are called 'closed systems'. Most of these networks were developed before standardization or when it was thought unlikely that equipment from other manufacturers would be included in the network.

In contrast, 'open systems' are those that conform to specifications and guidelines, which are 'open' to all. This enables equipment from various manufacturers, who claim compliance to a standard, to be used interchangeably on a network using that standard.

In 1978, faced with a proliferation of closed network systems, the International Standards Organization (ISO) defined a 'reference model for communication between

open systems'. This is known as the open systems interconnection (OSI) reference model, or more correctly as the ISO/OSI Model (ISO 7498). The model can be applied to all communication systems, from personal computers to satellite systems.

The ISO defines the purpose of the OSI reference model as:

'.... providing a common basis for the coordination of standards development for the purpose of systems interconnection, while allowing existing standards to be placed into perspective within the overall reference model.'

OSI is essentially a 'management structure', which simplifies data communications into a hierarchy of seven layers. Each layer has a defined purpose, which is dependent on, and interfaces with, the layer above and below it. Standards are defined for each layer in a way, which allows some flexibility enabling system designers to develop independent protocol layers. Any two or more of these layers together are referred to as a 'protocol stack'.

It is important to realize that the OSI reference model is not a protocol or set of rules dictating how a protocol should be written but an overall framework in which to define protocols. The OSI model framework specifically and clearly defines the functions or services that have to be provided at each of the seven layers (or levels).

The OSI reference model consists of the following seven layers:

• Layer 1, physical layer	Electrical and mechanical definition of the system
• Layer 2, data link layer	Framing and error correction format of data
• Layer 3, network layer	Optimum routing of messages from one network to another
• Layer 4, transport layer	Channel for transfer of messages of one application process to another
• Layer 5, session layer	Organization and synchronization of the data exchange
• Layer 6, presentation layer	Data format or representation
• Layer 7, application layer	File transfer, message exchange

The OSI model can be visualized as a collection of entities, such as software programs or hardware integrated circuits, situated at each of the seven layers. Data in a network is exchanged in packet form, each packet originating at a source node and addressed to a destination node. In effect, a packet starts at an upper layer, and passes down through each of the layers. As a packet moves down from one layer to another, it is enclosed in a 'protocol envelope'. Each envelope, therefore, encloses the message data and any protocol envelope from the layer above. The protocol envelope carries addressing and control information that advises the next layer down what to do with the packet, and the lower layer only reads that information.

When a packet reaches layer 1, that is the physical layer, it is sent across a physical communications link to the next node en route to its destination. At the receiving node, the packet moves up the protocol stack, losing the outer protocol envelope at each layer. In effect, the receiving node's network layer, for example, sees the packet as coming directly from the transmitting node's network layer because it has the appropriate envelope. For this reason, equivalent layers in different nodes are called 'peer entities' and there is said to be a 'virtual link' between them.

Figure 9.1 shows that a packet needs to go only as far as Layer 3, the network layer, on an intermediate node on its route. This is because the network layer protocol envelope contains all the information required for routing the packet along its journey.

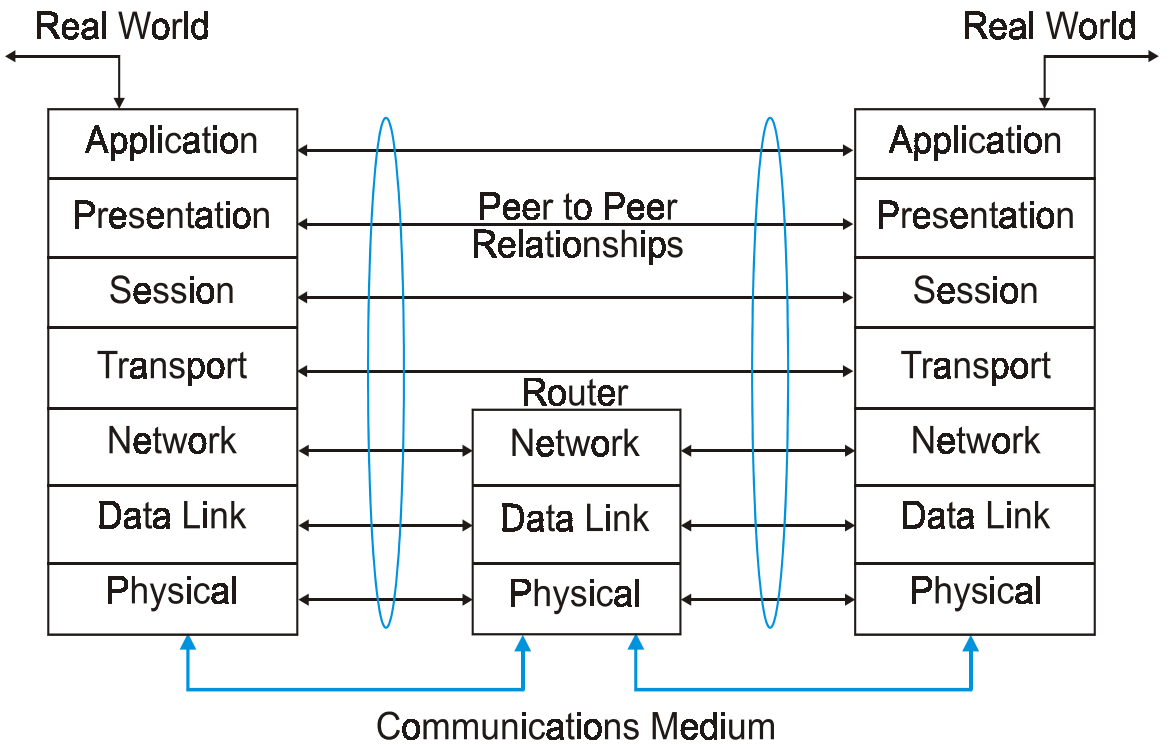


Figure 9.1
The layers of the OSI model

9.2 Individual OSI layers

Layer 7: The application layer

The application layer is the top most layer in the OSI/RM. It is responsible for giving applications access to the network. Examples of application layer tasks include file transfer, e-mail services and network management.

To accomplish the tasks, the application layer passes program requests and data to the presentation layer, which is responsible for encoding the application layer's data in the appropriate form.

Layer 6: The presentation layer

This layer maps the data representations into a suitable format for the recipient. It translates the format and syntax of the data produced by applications, and handles encryption or compression of the data.

Layer 5: The session layer

This layer manages interactive sessions between devices. It defines the use of software that allows reference to other devices by name rather than binary address (logical naming). It allows seamless recovery of sessions.

Layer 4: The transport layer

This layer defines:

- The management of the communications between two end systems.
- Data transfer at agreed levels of quality.
- How the packets in a large message are accounted for and correctly segmented. This improves the reliability of data transfer.

Layer 3: The network layer

This layer defines:

- How the packets of information are routed around the network.
- How the status messages are regulated and sent to other devices on the network.
- How large packets of data received from the transport layer are fragmented into small ones.
- Enough frames to go through the underlying network.

Layer 2: The data link layer

This layer is always needed and defines:

- The methods used to access the network for transmitting and receiving messages.
- The handling of the information received and its acknowledgment.
- The procedures for managing flow control between users.

Layer 1: The physical layer

This layer is always needed and defines the physical connections between the computer and the network. It provides Layer 2, the Data Link Layer, with the physical means of transmitting data onto the network.

This layer is concerned with the following:

- The network topology
- The electrical aspects of voltages and currents
- The signal modulation techniques
- The mechanical aspects of the connection, i.e. the cables and connectors

Note: It stipulates, but does not include the actual medium.

9.3 OSI analogy

The following is a simple analogy to better explain the OSI model.

A French-speaking manager in her Paris office wants to send a letter containing a request to an English-speaking colleague in an office in Chicago. The French manager

merely dictates the letter to her secretary who translates it into English. The French manager is like the application layer, while the secretary represents the presentation layer.

The letter is passed to a clerk who records its details in a file or correspondence with the Chicago company, making sure that details of title and address for the Chicago manager are correct. The clerk represents the session layer.

The shipping manager then receives the letter(s) and makes arrangements for the delivery and quality of service promised for the letter(s). He also makes a copy of the letter in case it has to be sent out again. He represents the transport layer.

Next in line is a shipping clerk who establishes the route for the letter. She decides that the best route for the letter is via the company's Boston office. She represents the network layer. The letter is then passed to the mailroom where it is weighed together with other letters going to Chicago by courier. The weight is labeled on the courier bag to ensure that there are no discrepancies at the other end. The mailroom is the data link layer.

The mailbag is then dropped off at the shipping dock, which acts as the physical layer.

Once the letter is received in Chicago the above process is repeated, but in reverse order.

9.4 An example of an industrial control application

The following is an example of the sequence of steps necessary to transfer a command, such as 'change the set point' on a remote programmable logic controller output:

- The user selects a command via the menu of a user program located on an operator station.
  For example, 'change the set point for a remote valve from 20% to 95%'.
- The menu-driven system software then instructs a command within the application layer program to make the necessary change. This layer can be visualized as being similar to the DOS interface on a computer, with a group of high level commands, acting as a high level interface.
- The application layer passes the message to the presentation layer, which translates it into a system usable form. For example, it could translate from the type coding used by the application layer into an ASCII based format suitable for the system.
- The session layer allows the two application protocol entities (at the operator station and the remote programmable logic controller) to synchronize, setup and manage the data interchange between them. This allows different requests to be queued and transmitted in an orderly manner.
- The transport layer forms the interface between the higher-level OSI layers and the lower network and data link layers. It shields the higher layers from the detailed operation of the lower layers. The transport layer provides various classes of service, for example adding the routing address onto the message received from the session layer.
- The message is passed onto the network layer, which arranges the detailed optimum routing, which the message is to follow across the network to arrive at the remote programmable logic controller.
- The message is then passed onto the data link layer, which calculates the frame check sequence for error checking and adds in source and destination addresses.
- The physical layer then modulates the physical connection with the appropriate bit pattern received from the data link layer.

The whole process is then repeated in reverse at the programmable logic controller.

9.5 Simplified OSI model

For many industrial protocols the use of the full seven layers of the OSI model is inappropriate as the application may require a high-speed response. Hence a simplified OSI model is often preferred for industrial applications where time critical communications is more important than full communications functionality provided by the seven-layer model.

Generally, most industrial protocols are written around three layers:

- The physical layer
- The data link layer
- The application layer

When the reduced OSI model is implemented the following limitations exist:

- As there is no transport layer, the maximum size of the application messages is limited by the maximum size allowed on the channel
- As there is no network layer, no routing of messages is possible between different networks
- As there is no session layer, no full duplex communications are possible
- As there is no presentation layer, message formats must be the same for all nodes

The MiniMap and Fieldbus protocol standards use the reduced three layer OSI model. Similarly, other industrial protocols such as the Allen Bradley Data Highway Plus protocol, Modbus Plus and the HART smart instrumentation protocols have all standardized on the three layers only.

One of the challenges when using the OSI model is the concept of interoperability and the need to define another layer called the ‘user’ Layer. This topic is examined in Chapter 12.